

Hypothesis Stress-Test & Investment Decision

Report v3.0.6.8

Powered by Vraimony

Protocol: ImmuneErrorRadar falsification protocol v3.0.6.9.7 | method=Vraimony methodology

| posture=standard_followed_not_copied

Falsification receipt: id=ierfp-bd7b3e911afb30f5bebc9cc |

contentSha256=bd7b3e911afb30f5bebc9cc | signedByVraimony=false

Documentation standard: ERF-inspired evidence receipt discipline: hash-first, portable, replayable, claim-bounded.

Citation line: This report follows the ImmuneErrorRadar falsification protocol, powered by Vraimony, based on the Vraimony evidence-receipt methodology.

Vraimony boundary: This receipt documents a falsification workflow and report payload hash; it is not medical advice, not clinical validation, not wet-lab evidence, and not proof of biology.

Report mode: verification-only | evidence-readiness receipt | investment-triage verdict | no discovery claim | no clinical advice

FINAL INVESTMENT DECISION v3.0.6.7: DO_NOT_INVEST_CLAIM_KILLED | claim=claim killed as written for current data |

killedClaim=HDAC_DNMT_epigenetic_mechanism_directly_drives_APM_loss_and_worse_PFS_in_this_dataset

| criteria=kill:4/invest:1 | next=Do not invest in HDAC/DNMT mechanism from this evidence.

Only APM data replication may be considered. | probabilityReported=false

PREPRINT READINESS v3.0.6.8: PREPRINT_READY_CLAIM_KILLED_AS_WRITTEN | title=Hypothesis

Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true |

residualDataFollowUp=false | wetLabAllowed=false | probabilityReported=false

Publication data fuel v3.0.6.9: route=HDAC_APM_FALLBACK |

state=HDAC_DNMT_MECHANISM_REOPEN_GATES_INJECTED_CLAIM_REMAINS_KILLED | syntheticData=false

| fakeReplication=false | defense=HDAC/DNMT mechanism remains killed because external

HDAC/MHC-I plausibility requires chromatin/perturbation/protein evidence not present in

current static bulk data. | APM multi-cohort fuel v3.0.6.9.6:

state=APM_MULTI_COHORT_DATA_FUEL_INJECTED_VALIDATION_TARGETS_ONLY | syntheticData=false |

fakeReplication=false | targets=medulloblastoma, Wilms tumor/nephroblastoma, Ewing sarcoma

| boundary=data-fuel only, not replication.

Pediatric cross-tumor scope v3.0.6.9.6: state=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE |

crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false

Required external pediatric tumors: medulloblastoma, Wilms tumor / nephroblastoma, Ewing sarcoma

Investment triage v3.0.6.6: verdict=KILL_HDAC_DNMT_MECHANISM_DO_NOT_INVEST |

killedClaim=HDAC_DNMT_epigenetic_mechanism_directly_drives_APM_loss_and_worse_PFS_in_this_dataset

| basis=kill criteria dominate and no defensible next investment exists for the supplied

claim; kill=4, invest=1 | next=Do not invest in HDAC/DNMT mechanism from this evidence.

Only APM data replication may be considered. | probabilityReported=false

Question understood: true

Question: In pediatric neuroblastoma GSE85047, HDAC-mediated epigenetic silencing

represses MHC-I antigen presentation genes (B2M, TAP1, HLA-A) and immune recognition

pathways. Tumors with high HDAC2 expression show concurrent low B2M/TAP1 and worse

progression-free survival independently of MYCN, reflecting a pharmacologically

reversible immune-invisible state driven by histone deacetylation.

Question understanding v3.0.3.97: HDAC_DNMT_APM_EPIGENETIC_REPRESSION | confidence=0.95
Mechanistic text understanding v3.0.4.3: HDAC_DNMT_EPIGENETIC_APM_REPRESSION_TEXT | confidence=0.96
Text source context: Google Co-Scientist/Nature epigenetic inhibition style text | prior=EPIGENETIC_APM_REPRESSION_SUPPORTED_GENERAL_MECHANISM
Text genes understood: B2M,HLA-A,HDAC2,TAP1,HDAC1,DNMT1
Mechanism claim parsed: HDAC/DNMT activity may silence antigen-presentation or immune-recognition programs; perturbation proof requires epigenetic/functional data.
Local proxy tests: HDAC2_APM_LOW_PROXY:PROXY_ONLY_NO_CHROMATIN_ASSAY | APM_LOW_FALLBACK:FALLBACK_DIRECT_EXPRESSION_SCORE
Not directly testable here: histone deacetylation at promoters,DNMT1 methylation layer,vorinostat/dual-inhibition perturbation effect,chromatin remodeling percentage
Why this route: HDAC2,histone deacetylation,epigenetic,B2M,TAP1
Cancer genes understood: B2M,TAP1,HLA-A,MYCN,HDAC2,HDAC1,DNMT1
Dataset/outcome resolved: GSE85047 | PFS | covariates=MYCN
Literature axis basis: Epigenetic repression text is decomposed into HDAC/DNMT upstream claim plus APM expression proxy.
Understanding caveats: epigenetic_silencing_requires_chromatin_or_perturbation_data
Hypothesis tested: HDAC2_DNMT1_APM_REPRESSION_PFS_MYCN_GSE85047
Primary axis: B2M_MHCL_NK_ESCAPE | route=PRIMARY_AXIS_HDAC_DNMT_APM_EPIGENETIC_TEXT_PROXY
Genes understood: B2M,TAP1,HLA-A,MYCN,HDAC2,HDAC1,DNMT1
Dataset/model: GSE85047 | outcome=PFS | model=HDAC/DNMT APM repression text proxy relaxed screen: APM-low fallback only because chromatin/perturbation data are not available
Initial result: INITIAL_EXPLORATORY_ADVERSE_SIGNAL_REVIEW_ONLY | priority=SECONDARY_LEAD_REVIEW_ONLY
Triage signal: REVIEW_ONLY_DIRECTIONAL_SCREEN | level=REVIEW_ONLY_DIRECTIONAL_SCREEN_CANDIDATE | verdict=REVIEW_ONLY_COMPUTATIONAL_ROUND_OFFICIAL_CLAIM_BLOCKED
Signal promotion: rawAllowed=true | forKilledClaimAllowed=false | residualDataFollowUpAllowed=false | nextComputationalRoundCandidate=true | wetLabCandidate=false | officialClaimAllowed=false
Directional decision v3.0.4.1: APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | local=PRIORITIZE_APM_SIGNAL_BUT_DO_NOT_ATTRIBUTE_TO_MECHANISM | rank=4
Best direction scan: APM_COMPOSITE_PFS_MYCN_GSE85047 | STRONG_EXPLORATORY_DIRECTION | score=12.3647
Directional ranking table: #1 APM composite:STRONG_EXPLORATORY_DIRECTION | #2 B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION | #3 CK2/APM fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #4 HDAC/APM fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #5 NK effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE | #6 $\gamma\delta$ /NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED | #7 CD276/B7-H3:SUBGROUP_ONLY_DIRECTION
Lead label: APM_FALLBACK_STRONG_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED
Directional why: Strong APM fallback signal is present, but it must not be attributed to the upstream mechanism without mechanism-specific probes or assay evidence: $\delta=0.191176$, rawHR=1.4159/p=0.0124, adjustedHR=1.1971/p=0.246.
Directional next action: Keep APM as the executed lead; acquire mechanism-specific probes/assays before attributing signal to CK2/HDAC or other upstream mechanisms.
Proxy attribution v3.0.4.4: FALLBACK_APM_PROXY_EXECUTED | signalSource=APM_COMPOSITE_FALLBACK | mechanismGeneTested=false |

label=APM_FALLBACK_STRONG_SIGNAL_EPIGENETIC_MECHANISM_NOT_DIRECTLY_TESTED
Probe lookup v3.0.5.0: resolvedVector=8 | resolvedButVectorMissing=20 | probeUnknown=0 |
absentOrReporterUnavailable=2 |
staticManifestLocalResolve=TARGETED_STATIC_MANIFEST_ATTACHED | runtimeNetwork=false |
legacyAutoFetchFlag=true
Probe vector binding v3.0.5.1: columnIndexPresent=true | resolvedProbelds=28 |
boundVectors=8 | resolvedButColumnMissing=20 | aliasBoundVectors=0 |
nextBlocker=PARTIAL_BINDING_REMAINING_GENES_NEED_COLUMN_EXTRACTION
Evidence fuel v3.0.6.0: HDAC_APM_FALLBACK |
APM_FALLBACK_FUEL_AVAILABLE_HDAC_MECHANISM_NOT_BOUND |
expected=APM_FALLBACK_SIGNAL_MECHANISM_NOT_ATTRIBUTABLE |
boundary=DATA_FUEL_IS_TEST_RECIPE_AND_SELECTED_RUNTIME_COVERAGE_NOT_TIER2_PROOF
Executed local directions v3.0.4.4: #1 APM
composite:STRONG_EXPLORATORY_DIRECTION[APM_COMPOSITE_DIRECT_PROXY] | #2
B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION[B2M_TAP1_NK_DIRECT_PROXY] | #5 NK
effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY]
| #6
γδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED[GAMMA_DELTA_S
| #7 CD276/B7-H3:SUBGROUP_ONLY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY] |
#8
ODC1/polyamine:WEAK_EXPLORATORY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY]
Mechanistic hypotheses needing assets v3.0.4.4: CK2/APM
fallback:APM_FALLBACK_STRONG_SIGNAL_CK2_MECHANISM_NOT_DIRECTLY_TESTED | HDAC/APM
fallback:APM_FALLBACK_STRONG_SIGNAL_EPIGENETIC_MECHANISM_NOT_DIRECTLY_TESTED |
γδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED |
NECTIN2/TIGIT:NECTIN2_TIGIT_NOT_TESTABLE_PROBE_AND_SINGLE_CELL_CONTEXT_REQUIRED
N=272 | events=96 | highEventRate=0.448529 | lowEventRate=0.257353 | delta=0.191176
Cox first-pass: state=RELAXED_COX_FIRST_PASS_EXECUTED_REVIEW_ONLY | HR=1.4159 | p=0.0124 |
rows=272 | events=96
MYCN-adjusted Cox first-pass: state=RELAXED_COX_FIRST_PASS_EXECUTED_REVIEW_ONLY |
HR=1.1971 | p=0.246 | rows=272 | events=96
MYCN-stratified event-rate: nonMYCN delta=0.077621 n=218 | MYCN-amp delta=0.242026 n=54
Component separation: antigenLoss delta=0.191176 | nkLow delta=0.352941 | combined
delta=0.191176
Tier-1 package v3.0.4.2: TIER1_FULL_GPL5175_LOCAL_COX_TARGET |
current=TIER0_DIRECTIONAL_SELECTED_MATRIX_ACTIVE | noStrictnessIncrease=true
GPL5175 annotation: FULL_GPL5175_ANNOTATION_REQUIRED_BEFORE_TIER1_PASS | fullPass=false
Verified-local Cox package: VERIFIED_LOCAL_COX_RERUN_CANDIDATE_RELAXED_MODE |
label=SECONDARY_AXIS_VERIFIED_LOCAL_COX_RERUN_TARGET_APM_FALLBACK_SIGNAL_MECHANISM_NOT_D
Reproducibility receipt:
sha256-4c7a5c79e5a0556c1b9ba36554ced900fea294080580b76197be2336fc44a066 |
replay=REPLAY_PACKAGE_SPEC_READY_CURRENT_RUN_EMBEDDED_JSON_ONLY
Method claim: The app parses cancer hypotheses, maps them to expression-survival axes,
runs relaxed first-pass event-rate/Cox screens, ranks directional signals, and packages
reproducibility receipts while blocking official proof/kill claims before Tier-2.
Biology claim: This axis is interpreted only as a local directional result in the current
selected-matrix screen.
Evidence readiness gate: BLOCKED_BY_MISSING_EVIDENCE |
missing=official_or_declared_probe_platform_mapping,probe_to_gene_mapping_receipt,independent_replication_o
Evidence readiness receipts: platform/probe=PARTIAL_MAPPING_REVIEW_ONLY |

outcomeResolved=true | covariateResolved=true |
 phenotypeComparator=SURVIVAL_OUTCOME_MODEL_NO_PHENOTYPE_COMPARATOR_REQUIRED_OR_NOT_DECLARED |
 replicationComparator=REPLICATION_COMPARATOR_NOT_DECLARED |
 confounder=CONFOUNDER_ADJUSTED_MODEL_EXECUTED_REVIEW_ONLY | mycnCoxExecuted=true |
 fragility=RAW_SIGNAL_STRONG_BUT_WEAKENED_AFTER_MYCN_ADJUSTMENT_REVIEW_ONLY
 Axis continuation: AXIS_FOLLOW_UP_REQUIRED_NOT_NEW_DISCOVERY | next=Do not stop at
 promising. Run axis transfer, direction-stability, known-driver and bulk-attribution
 checks using the supplied axis only.
 Operational readiness: BLOCKED_FROM_PROMOTION_BY_READINESS_GAPS — Resolve first:
 official_or_declared_probe_platform_mapping, probe_to_gene_mapping_receipt,
 independent_replication_or_cohort_transfer_check
 Final hypothesis state v3.0.6.0: APM_FALLBACK_SIGNAL_MECHANISM_NOT_ATTRIBUTABLE |
 reason=APM-low direction may be tested locally, but HDAC/DNMT epigenetic mechanism is not
 directly attributable without chromatin, perturbation, or mechanism-specific vector
 evidence.
 Decisive adjudication v3.0.6.5:
 verdict=KILLED_HDAC_DNMT_MECHANISM_ATTRIBUTION_APM_LOCAL_DIRECTION_SURVIVES |
 claimKilled=true | localSignalSurvives=true | notTestable=false | action=Stop mechanism
 claims; keep only the local APM direction for replication or full mapping.
 Investment verdict v3.0.6.6: verdict=KILL_HDAC_DNMT_MECHANISM_DO_NOT_INVEST |
 killedClaim=HDAC_DNMT_epigenetic_mechanism_directly_drives_APM_loss_and_worse_PFS_in_this_dataset
 | killCriteria=4 | investCriteria=1 | next=Do not invest in HDAC/DNMT mechanism from this
 evidence. Only APM data replication may be considered.
 FINAL INVESTMENT DECISION v3.0.6.7: DO_NOT_INVEST_CLAIM_KILLED | claim=claim killed as
 written for current data |
 killedClaim=HDAC_DNMT_epigenetic_mechanism_directly_drives_APM_loss_and_worse_PFS_in_this_dataset
 | criteria=kill:4/invest:1 | next=Do not invest in HDAC/DNMT mechanism from this evidence.
 Only APM data replication may be considered. | probabilityReported=false
 PREPRINT READINESS v3.0.6.8: PREPRINT_READY_CLAIM_KILLED_AS_WRITTEN | title=Hypothesis
 Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true |
 residualDataFollowUp=false | wetLabAllowed=false | probabilityReported=false
 Publication data fuel v3.0.6.9: route=HDAC_APM_FALLBACK |
 state=HDAC_DNMT_MECHANISM_REOPEN_GATES_INJECTED_CLAIM_REMAINS_KILLED | syntheticData=false
 | fakeReplication=false | defense=HDAC/DNMT mechanism remains killed because external
 HDAC/MHC-I plausibility requires chromatin/perturbation/protein evidence not present in
 current static bulk data. | APM multi-cohort fuel v3.0.6.9.6:
 state=APM_MULTI_COHORT_DATA_FUEL_INJECTED_VALIDATION_TARGETS_ONLY | syntheticData=false |
 fakeReplication=false | targets=medulloblastoma, Wilms tumor/nephroblastoma, Ewing sarcoma
 | boundary=data-fuel only, not replication.
 Pediatric cross-tumor scope v3.0.6.9.6: current=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE |
 observed=neuroblastoma | crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false
 Pediatric expansion targets: medulloblastoma, Wilms tumor / nephroblastoma, Ewing sarcoma
 Pediatric scope next action: After neuroblastoma, run the same stress-test template on
 medulloblastoma, Wilms tumor/nephroblastoma, and Ewing sarcoma; keep each as KILLED /
 NULL-LIKE / SURVIVING / NOT_MEASURABLE with separate receipts before any cross-tumor
 claim.
 Pediatric scope boundary: Neuroblastoma-only evidence remains single-cohort and
 disease-scoped. Cross-pediatric-tumor credibility requires independent human datasets from
 at least three additional pediatric tumor families before any generalization, Tier-2
 promotion, or wet-lab language.

Preprint boundary: This report supports a methods/resource-allocation claim only: supplied computational hypotheses are triaged as killed-as-written, watchlist-only, or data-only follow-up. It does not prove or disprove biology, does not recommend treatment, and does not establish clinical actionability.

Allowed manuscript sentence: The supplied claim is killed as written for the current data; the result should redirect resources away from the claimed mechanism rather than imply biological impossibility.

Direction: HIGH_SCORE_WORSE_PFS_EVENT_RATE | caveat=Relaxed first-pass screen only. Shows trial event-rate, MYCN-stratified and in-app Cox-review numbers for debugging. Not proof, not official support/kill, not clinical, not publication-grade, and not Tier-2 computational evidence.

Missing gates: official HDAC2/DNMT1 full GPL5175 mapping,chromatin/HDAC perturbation not available,DNMT methylation layer not available,full GPL5175 audit,source certification,independent Cox-capable replication,Tier-2 not reached,full GPL5175 mapping required before official support/kill,independent replication required before computational evidence language

Claim guard: BLOCKED_COMPUTATIONAL_CLAIM_REQUIRES_TIER2 | tier=TIER0_SLICE_REPRODUCIBLE_REVIEW_ONLY | officialPromotionAllowed=false | experimentalPromotionAllowed=false | experimentalPromotionForKilledClaimAllowed=false | dataFollowUpAllowed=false | computationalReplicationAllowed=false | wetLabAllowed=false | officialClaimAllowed=false

Tier-0 proves replayability, not source truth.

Tier-1 proves source extraction, not biological truth.

Tier-2 is required before any computational evidence claim.